

Exploratory Data Analysis (EDA)

Executive Summary

1. Problem Statement

The primary objective of this analysis is to perform an exploratory investigation of the sales dataset to uncover underlying patterns, understand customer purchasing behavior, and identify any anomalies or outliers prior to further analysis.

2. Methodology

Data Preparation: Utilized the previously cleaned dataset, where missing values were handled and date formats were standardized.

Analysis Tool: Employed Power BI for descriptive statistics and visual diagnostics.

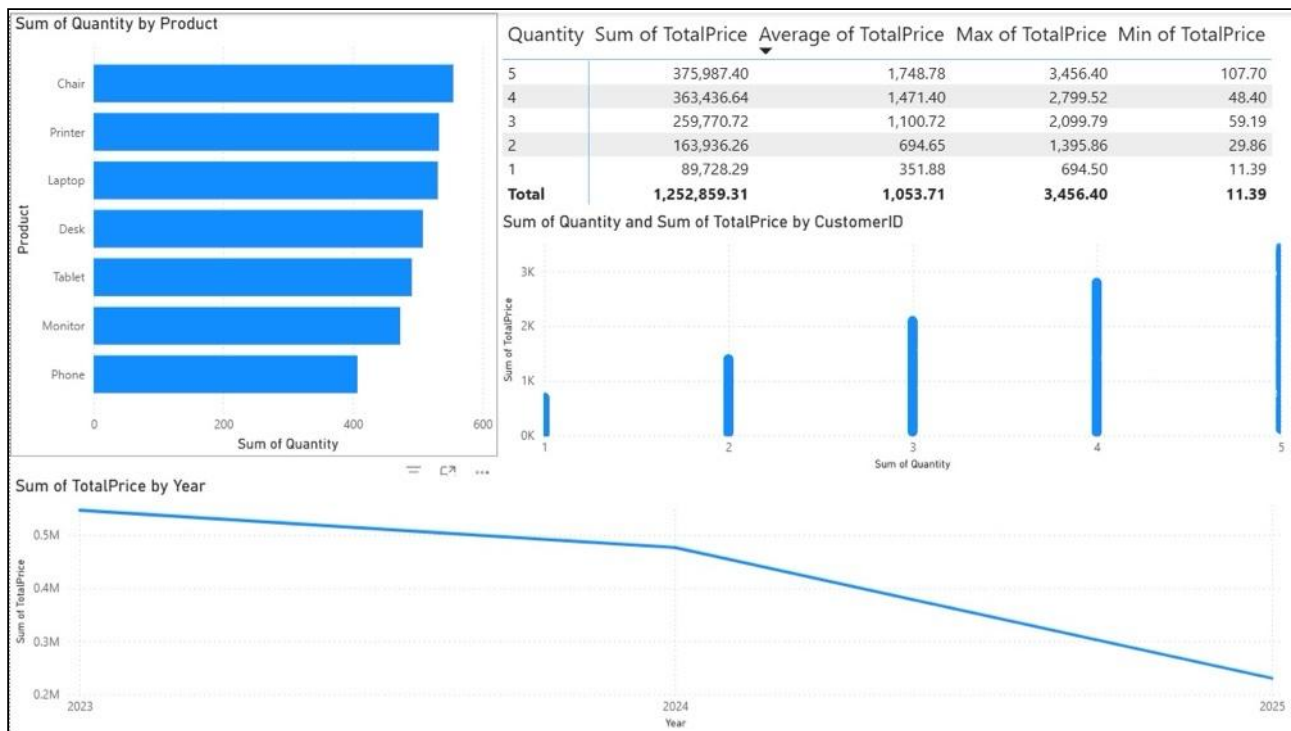
Techniques Applied: Extracted the five-number summary, evaluated data distribution, and utilized scatter, line, and bar charts for trend and outlier detection.

3. Key Findings

Descriptive Statistics: The average order value (Total Price) is \$1,053.71, with a minimum of \$11.39 and a maximum of \$3,456.40.

Product Performance: "Chair" is the highest-selling product by quantity, while "Phone" records the lowest sales volume.

Outlier Detection: Scatter chart analysis correlating Quantity and Total Price revealed that pricing scales logically with quantity. No extreme or erroneous outliers were detected in the transactions.



4. Recommendations

Investigate Low Performers: Analyze the root causes behind the low sales volume of "Phones" (e.g., pricing strategy, market competition, or marketing reach).

Capitalize on High Demand: Leverage the high demand for "Chairs" by introducing complementary products (like desks) or bundled promotional offers.

Continuous Monitoring: Maintain regular checks on transaction volumes and order IDs to ensure ongoing data integrity and catch any future anomalies early.